Spatial resource arrangement influences both network structures and activity of fungal mycelia: A form of pattern recognition?

Yu Fukasawa; Kosuke Hamano; Koji Kaga; Daisuke Akai; Takayuki Takehi

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Abstract

The present study investigated the behavior and wood decay ability of mycelial network of *Phanerochaete velutina*, a cord-forming fungus, on multiple wood blocks. We placed well-colonized wood blocks in two spatial arrangements (Circle and Cross) on a soil plate and compared the development of the mycelial network and wood decay over 116 days. In the Circle arrangement, the degree of connection (number of connected cords) of the blocks ranged from 0 to 8, with no significant differences observed across positions. However, in the Cross arrangement, the outer blocks exhibited a greater degree of connection than the inner blocks. The mass loss of the wood block was positively associated with the degree of connection and was significantly smaller in the Cross than in the Circle arrangement. These findings suggest that fungal mycelium can “recognize” the difference in the spatial arrangement of wood blocks as part of their wood decay activity.